

INTRODUCTION

Supply Chain Management (SCM) is an essential part of every organization that deals with the production, movement, and delivery of goods and services. It involves the coordinated management of sourcing, manufacturing, transportation, and distribution processes to ensure that products reach customers efficiently and at the right cost. A well-designed supply chain enhances overall business productivity and plays a major role in meeting customer demands, improving quality, and ensuring timely delivery.

In today's competitive and globalized marketplace, companies face increasing challenges such as rising costs, market fluctuations, and changing consumer expectations. Effective supply chain management helps businesses stay agile by optimizing operations, minimizing waste, and improving cost efficiency. It acts as a bridge connecting suppliers, manufacturers, and consumers, ensuring smooth communication and collaboration across every stage of the production cycle.

The importance of SCM has grown even more with the integration of technology and data analytics. Modern businesses now rely heavily on analytical tools to monitor performance, forecast demand, and evaluate operational efficiency. Through real-time data visualization, organizations can identify performance gaps, track inventory movement, and make strategic decisions to improve productivity and profitability.

Therefore, understanding and analyzing the supply chain process is vital for continuous improvement and strategic decision-making. This project, "Supply Chain Management Dashboard: Power BI," is developed to visualize and analyze various aspects of the supply chain such as logistics efficiency, production cost, and revenue generation.

ABSTRACT

The project titled “Supply Chain Management Dashboard: Power BI” focuses on analyzing supply chain performance through interactive data visualization. Using Microsoft Power BI, the project collects, processes, and visualizes key business metrics to evaluate how efficiently manufacturing, logistics, and product categories perform. It aims to transform raw operational data into clear and meaningful insights that support better business decisions and improved efficiency.

The dataset used in this project includes multiple factors such as total revenue, cost per unit, defect rates, transport modes, and manufacturing costs. These variables are analyzed to identify trends, measure cost efficiency, and assess product performance. The use of KPI cards, bar charts, pie charts, and scatter plots enables the dashboard to display complex data in an easily understandable format, making it accessible for business analysts and decision-makers.

The dashboard highlights key metrics such as Total Revenue (578K) and Average Cost per Unit (0.11), along with insights into logistics share and product quality. The analysis reveals that while road transport dominates logistics, it also presents cost fluctuations. Meanwhile, air and rail transport show better consistency in specific product categories. Defect rates and cost variations further indicate opportunities for process optimization in manufacturing and distribution.

Overall, the project demonstrates how Power BI can be used as a powerful tool for monitoring, analyzing, and improving supply chain operations. By visualizing key data points and enabling interactive filtering, the dashboard allows organizations to identify inefficiencies, reduce costs, and enhance overall supply chain performance.

OBJECTIVES

- To analyze overall supply chain performance by assessing total revenue generation, cost per unit, and product category contribution.
- To evaluate logistics efficiency by comparing different transport modes (Air, Rail, Road, and Sea) and identifying cost-effective and time-efficient options.
- To examine manufacturing performance through an analysis of production cost, defect rates, and quality outcomes across various product categories.
- To identify trends and patterns in cost efficiency, order volumes, and defect distribution that can influence supply chain decision-making.
- To utilize Power BI for interactive data visualization, enabling users to filter, explore, and interpret key metrics dynamically for better operational understanding.
- To establish data-driven KPIs that provide a clear overview of financial and operational performance within the supply chain.
- To highlight areas of improvement in production planning, transportation management, and quality control for enhanced overall efficiency.

DATA ANALYSIS

PROCEDURE

1. Data Collection

Relevant supply chain data was collected from company records and logistics management systems. The dataset includes information on revenue generation, transportation modes, manufacturing costs, order quantities, defect rates, and cost per unit. The data was gathered to analyze supply chain performance, logistics efficiency, and product category trends.

2. Data Cleaning and Preprocessing

The raw dataset was carefully reviewed to remove missing or duplicate values and to correct inconsistencies in entries such as transport type or product category. Data was categorized into numerical (e.g., cost, revenue, defect rates) and categorical variables (e.g., transport mode, product type) to ensure smooth analysis and visualization.

3. Data Organization

The cleaned dataset was structured into key dimensions and measures:

- **Dimensions:** Product Type, Transport Mode, Manufacturing Unit.

- **Measures:** Revenue, Cost per Unit, Defect Rates, Order Quantities.

This step ensured consistency and comparability between different data components for deeper insights.

4. Exploratory Data Analysis (EDA)

Descriptive analytics were conducted to summarize and understand the dataset:

- Total revenue generation
- Cost distribution per unit
- Manufacturing and defect patterns Correlation analysis was done to identify relationships between transport mode, cost efficiency, and product quality.

5. Data Visualization

Using Microsoft Power BI, interactive dashboards were designed to represent supply chain performance visually.

The following key visuals were used:

- **KPI Cards:** To highlight total revenue and average cost per unit.
- **Pie Charts:** To show the share of logistics by transport mode.
- **Bar/Column Charts:** To compare revenue, orders, and defect rates by product category.

- **Scatter Plot:** To analyze manufacturing cost efficiency against production volumes.

6. Interpretation of Results

The dashboard provided clear insights into cost efficiency, logistics performance, and product category profitability.

Key patterns included:

- Certain product categories (like stationary) generated higher revenue but had higher defect rates.
- Road transport accounted for the majority logistics share, while air and rail contributed marginally.
- A noticeable trade-off between production volume and cost efficiency was identified.
- Products with higher order quantities did not always yield proportionally higher revenue, indicating optimization opportunities.

7. Conclusion from Analysis

The analysis revealed multiple interrelated factors affecting supply chain performance.

Key takeaways:

- Transportation cost and mode significantly influence overall profitability.
- Defect management and manufacturing optimization are crucial for improving efficiency.

- Product-wise revenue contribution helps in resource allocation and demand forecasting.
- Visual dashboards allow stakeholders to monitor KPIs and take data-driven decisions for cost reduction and process improvement.

ABOUT DATASET

The dataset used for this project comprises logistics, production, and sales data from various product categories and transport modes.

Key Attributes in the Dataset:

- **Revenue Generated:** Total income from each product category.
- **Cost per Unit:** Average manufacturing or logistics cost.
- **Transport Mode:** Categories such as Air, Rail, Road, and Sea.
- **Manufacturing Costs:** Total production costs per product batch.
- **Order Quantities:** Number of units ordered by category.
- **Defect Rates:** Quality performance measure based on defective items.
- **Product Category:** Main dimension — Stationary, Cosmetics, Electrical, etc.

Data Purpose

- To evaluate overall revenue and cost efficiency.
- To identify logistics performance by transport mode.
- To analyze production cost efficiency across product types.
- To assess quality performance based on defect rates.
- To uncover optimization opportunities in manufacturing and logistics.

METHODOLOGY

Tool Used

- **Microsoft Power BI**

Used for data import, transformation, modeling, visualization, and dashboard creation.

Visualizations Used

- **KPI Cards:**

- Total Revenue (578K)
- Average Cost per Unit (0.11)

- **Pie Chart:**

- Logistics share by transport mode (Air, Rail, Road, Sea)

- **Bar Charts / Column Charts:**

- Revenue contribution by product category
- Total orders by product category
- Quality performance by product category

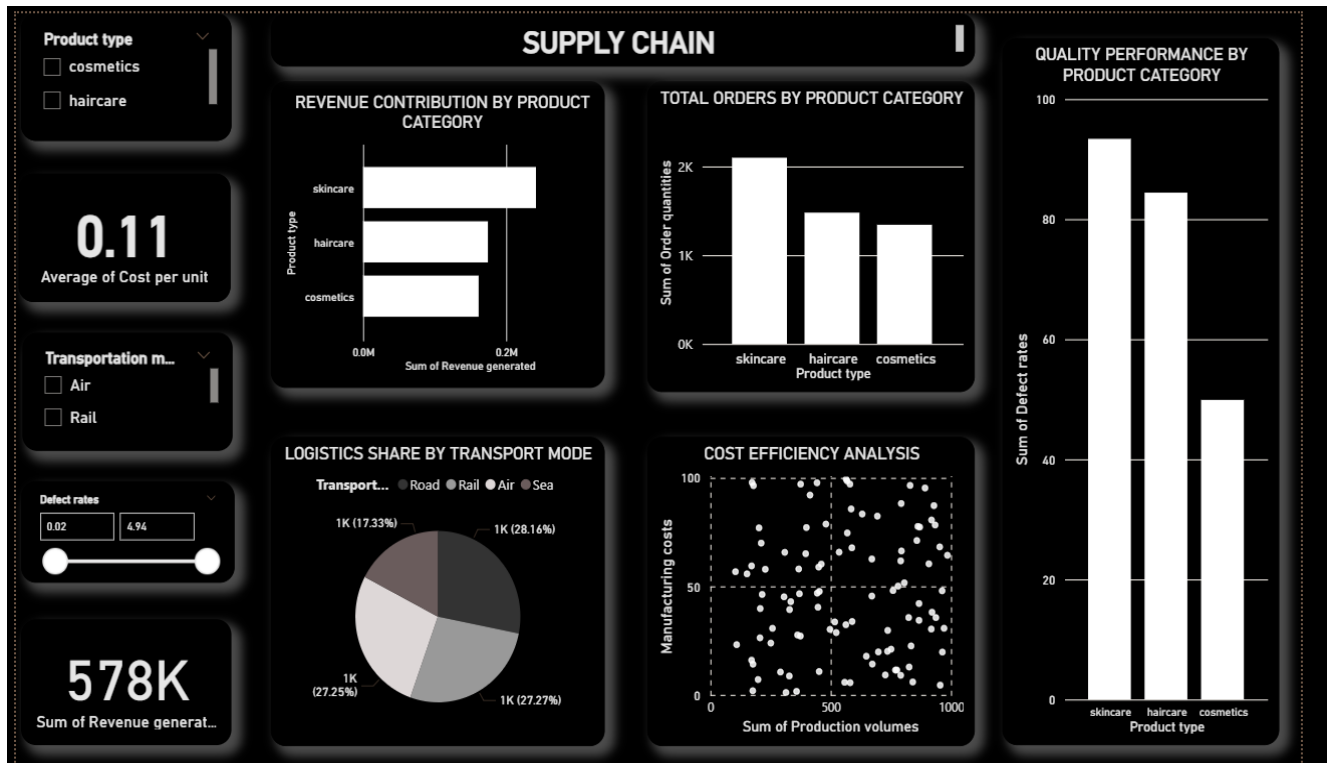
- **Scatter Plot:**

- Cost efficiency analysis (Manufacturing Cost vs. Production Volume)

Interactive Components

- **Slicers:** For filtering by product category or transport mode.
- **Tooltips:** For showing detailed metrics on hover.
- **Drill-Through Pages:** To navigate from overall KPIs to detailed category-level insights.

DATA VISUALIZATION:



DATA INTERPRETATION

Dependent Variables (What is Measured)

- **Revenue Generation:** Sum of revenue across all products.
- **Cost Efficiency:** Average cost per unit and manufacturing cost analysis.
- **Quality Performance:** Sum of defect rates across product types.

Independent Variables (What Influences the Results)

- **Transport Mode:** Impacts logistics cost and delivery efficiency.
- **Product Type:** Determines cost structure, defect rate, and profitability.
- **Manufacturing Volume:** Affects cost efficiency and defect probability.

Key Insights

- **Financial Efficiency:**
 - Road transport dominates logistics share but increases cost variability.
 - Lower transport modes (Air/Rail) have smaller shares but better delivery precision.
- **Quality Trends:**
 - Stationary products recorded the highest defect rates.

- Cosmetics performed better in quality and cost consistency.
- **Operational Optimization:**
 - Scatter plot analysis shows some manufacturing units operating above optimal cost efficiency levels.
 - Aligning production volumes with demand can reduce waste and defects.

KEY PERFORMANCE INDICATORS (KPIs)

The Supply Chain Management dashboard highlights the following key performance indicators that provide a clear overview of business performance and operational efficiency:

- **Total Revenue Generated (578K):**
Indicates the overall income generated across all product categories.
- **Average Cost per Unit (0.11):**
Reflects the average production or logistics cost incurred per unit of product.
- **Logistics Share by Transport Mode (%):**
Displays the proportional distribution of shipments via Air, Rail, Road, and Sea transportation modes.
- **Revenue Contribution by Product Category (%):**
Shows how each product type (e.g., Stationary, Electrical, Cosmetics) contributes to the total revenue.
- **Total Orders by Product Category:**
Represents the total quantity of orders placed for each category, identifying high-demand products.
- **Manufacturing Cost Efficiency:**
Visualizes the relationship between production volume and manufacturing cost to assess operational efficiency.

- **Defect Rate (%):**

Indicates the proportion of defective units in total production, a critical measure of quality performance.

- **Quality Performance by Product Category:**

Compares defect rates across product types to evaluate manufacturing reliability.

- **Transport Cost Comparison:**

Measures and compares cost efficiency across different logistics modes.

SLICERS

To enable interactive and dynamic analysis, the dashboard incorporates several slicers that allow users to filter and explore data based on multiple dimensions:

- **Transport Mode:**

Enables filtering by logistics type — Air, Rail, Road, Sea — to compare transportation efficiency and cost.

- **Product Category:**

Allows users to analyze metrics (revenue, defects, orders) for specific categories such as Cosmetics, Stationary, or Electrical.

- **Manufacturing Unit:**

Filters performance data based on individual production plants or facilities.

- **Revenue Range:**

Enables selection of data based on different revenue brackets for focused financial analysis.

- **Cost per Unit Range:**

Allows narrowing down products based on cost efficiency or high-cost items.

PROJECT INSIGHTS – SUPPLY CHAIN MANAGEMENT ANALYSIS REPORT

- The total revenue of 578K indicates a strong overall performance, yet the average cost per unit (0.11) suggests potential for further cost optimization.
- Road transport contributes the largest share of logistics activity, emphasizing its importance but also the need for efficient route and cost management.
- Air and Rail transport modes have relatively lower shares but demonstrate better efficiency in cost and delivery time for specific product categories.
- Revenue contribution analysis shows that certain product types—such as stationary and electrical—generate higher income but may also incur higher defect rates.
- The manufacturing cost efficiency scatter plot reveals that not all production units operate at optimal cost-to-volume ratios, indicating opportunities for better process control.
- Defect rate analysis highlights quality challenges in certain product lines, especially in high-volume categories, emphasizing the need for improved quality control systems.
- The total orders vs. revenue comparison shows that higher order volumes do not always translate to proportionally higher profits, suggesting inefficiencies in pricing or production planning.

- Cost efficiency trends reveal that minimizing production waste and enhancing process automation can significantly improve profitability.
- Insights from transportation mode comparisons underline the importance of balancing cost, delivery time, and reliability when planning logistics operations.
- The findings collectively emphasize the critical role of data-driven decision-making in optimizing cost, logistics, and quality across the supply chain.

CONCLUSION

The analysis underscores that supply chain performance is shaped by a complex interaction of logistics efficiency, manufacturing cost control, product quality, and transportation decisions.

Key performance indicators such as total revenue (578K), average cost per unit (0.11), and defect rates provide actionable metrics for evaluating operational health.

The dashboard reveals that while overall revenue is healthy, manufacturing inefficiencies, defect rates, and logistics costs are significant areas requiring attention.

Optimizing transport routes, balancing production volumes, and improving quality control mechanisms can lead to substantial cost savings and improved profitability.

By leveraging Power BI visual analytics, organizations can monitor performance in real time, identify process bottlenecks, and make informed, data-driven decisions.

This analysis highlights the importance of continuous monitoring, proactive strategy, and cross-functional coordination to strengthen overall supply chain resilience and competitiveness.

SUGGESTIONS

1. Optimize Transportation Efficiency

- **Adopt Route Optimization Tools:** Use predictive analytics to reduce fuel and time costs for road transport.
- **Balance Mode Selection:** Evaluate the trade-offs between air, rail, and road modes to improve delivery efficiency and cost balance.

2. Improve Manufacturing Cost Efficiency

- **Implement Lean Production Practices:** Eliminate process waste and improve production flow to lower manufacturing costs.
- **Monitor Production KPIs Regularly:** Track cost per unit and output volume to identify and correct inefficiencies early.

3. Strengthen Quality Control Systems

- **Introduce Real-Time Quality Checks:** Deploy automated quality monitoring tools to minimize defect rates.
- **Train Workforce on Quality Standards:** Conduct regular workshops to ensure uniform production and packaging quality.

4. Enhance Data-Driven Decision-Making

- **Utilize Dashboard Insights:** Regularly review Power BI dashboards to track key metrics and detect anomalies.

- **Enable Cross-Departmental Access:** Allow logistics, finance, and production teams to collaborate using the same data platform.

5. Refine Product Category Strategy

- **Focus on High-Revenue, Low-Defect Categories:** Prioritize product lines that demonstrate strong profitability with minimal quality issues.
- **Reassess Low-Performing Categories:** Identify underperforming products and revise manufacturing or marketing strategies accordingly.

6. Reduce Logistics and Storage Costs

- **Adopt Just-in-Time Inventory:** Minimize storage costs and reduce overproduction risks.
- **Negotiate with Transport Vendors:** Secure long-term contracts or partnerships for better freight rates and reliability.

7. Continuous Performance Improvement

- **Establish Monthly KPI Reviews:** Schedule regular performance evaluations to track progress against targets.
- **Encourage Feedback Loops:** Involve employees and managers in continuous process improvement initiatives to sustain operational excellence.